AIS Tracking Challenge Dataset

13 Scenario Types · 20 Geographic Variants Each · 39,340 Annotated Pings

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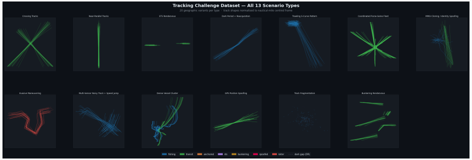
Activity Intelligence Engine

April 2026

What's In This Package

Purpose

A reproducible benchmark dataset for evaluating AIS data-association and activity-classification algorithms against the hardest real-world tracking problems.



Contents

- **13 scenario types** — each a canonically difficult tracking problem
- **20 variants per type** — different geographic regions, RNG seeds, and speed scales
- **260 CSV files** + 13 merged per-type files + 1 master flat file

Dataset Schema

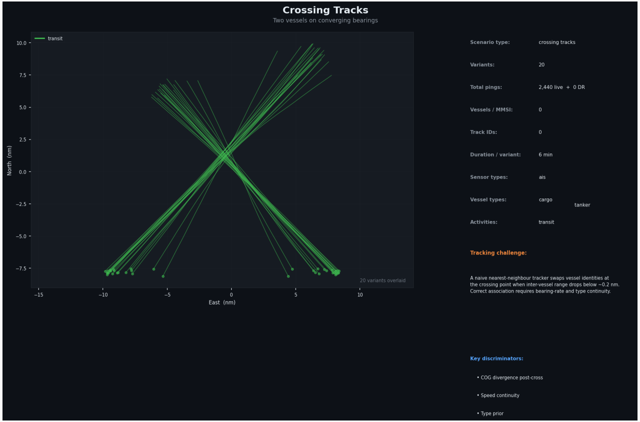
Every row in every CSV shares the same 18-column schema:

Column	Description
`mmsi`	Vessel MMSI — may repeat across tracks (clone/spoofing scenarios)
`timestamp`	ISO 8601 UTC — sort ascending for sequential replay
`lat`, `lon`	WGS-84 decimal degrees
`sog`	Speed over ground, knots (NaN during dark-gap rows)
`cog`, `heading`	Course/heading, degrees (NaN during dark-gap rows)
`sensor_type`	`ais` — `satellite_ais` — `radar` — `eo` — `none`
`true_activity`	Ground-truth label: *fishing*, *transit*, *anchored*, *sts*, *bunkering*, *spoofed*, *loiter*
`vessel_type`	*cargo*, *tanker*, *fishing*, *trawler*, *purse_seiner*, . . .
`nav_status`	AIS navigation status integer code
`scenario`	Canonical scenario key
`track_id`	Per-scenario track identifier (e.g. `A`, `clone`, `burst_3`)
`is_dark`	`True` for AIS-gap / dead-reckoning interpolated rows
`variant_id`	1–20; identifies the geographic/parametric variant

Scenario Index

#	Scenario	Core challenge
1	Crossing Tracks	ID swap at crossing; nearest-neighbour failure
2	Near-Parallel Tracks	<0.3 nm separation, identical COG/SOG
3	STS Rendezvous	Converge $\rightarrow$ 60 min dwell $\rightarrow$ diverge
4	Dark Reacquisition	4 h AIS gap; reappears off dead-reckoned position
5	Trawling Pattern	S-curve hauls at 3 kn $\rightarrow$ 10 kn transit transition
6	Coordinated Fleet	6 purse seiners in ≤ 0.2 nm orbit
7	MMSI Clone	Same MMSI from two positions 15 nm apart
8	Evasive Maneuvering	Turn rates $>20^\circ$/min; breaks CV prediction
9	Noisy Multi-Sensor Track	AIS gaps + EO outliers + 3-phase speed jump
10	Dense Cluster	12 vessels in 0.5 nm radius
11	Position Spoofing	Frozen GPS broadcast vs. satellite truth
12	Track Fragmentation	6 bursts (3–7 pings) with 20–60 min gaps
13	Bunkering Rendezvous	Near-stationary pair triggers false STS alert

1 · Crossing Tracks



What happens

Two vessels cross at near-right angles. At closest approach (≈0.2 nm), a nearest-neighbour tracker swaps identities.

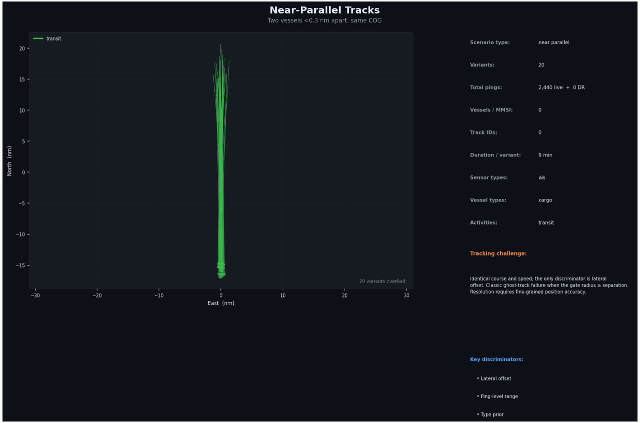
Why it's hard

Proximity-based gating fails when range temporarily drops below gate radius. Bearing-rate and vessel-type continuity are required to resolve the swap.

Ground truth

`track_id=A` and `track_id=B`

2 · Near-Parallel Tracks



What happens

Two cargo vessels travel north at 10 kn, separated by only $\sim$0.25 nm ($\approx$450 m). Courses and speeds are nearly identical.

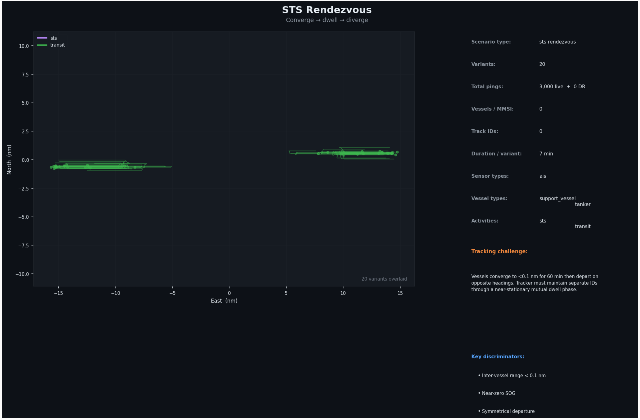
Why it's hard

Any tracker with a gate radius $\geq$0.25 nm conflates the two tracks into one (ghost-track). Fine-grained position accuracy is the only separator.

Ground truth

Both vessels maintain

3 · STS Rendezvous



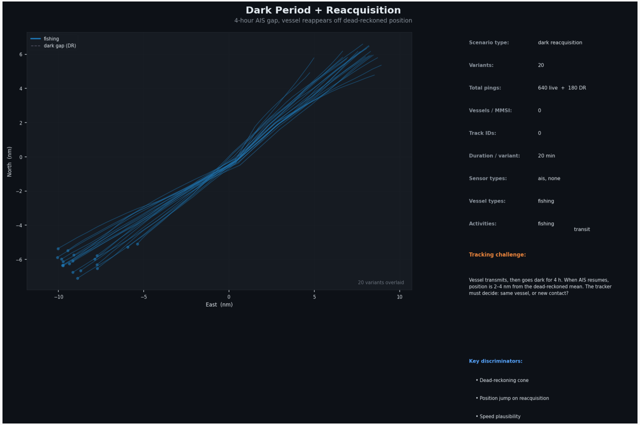
What happens

Tanker and support vessel converge over 45 min, hold position together for 60 min (STS transfer), then depart on opposite headings.

Why it's hard

During the dwell phase both vessels are near-stationary and <0.1 nm apart. The tracker must maintain distinct IDs through overlap and correctly assign post-departure tracks.

4 · Dark Reacquisition



What happens

Fishing vessel transmits for 90 min, goes AIS-dark for 4 h, then reappears 2–4 nm from the dead-reckoned mean position.

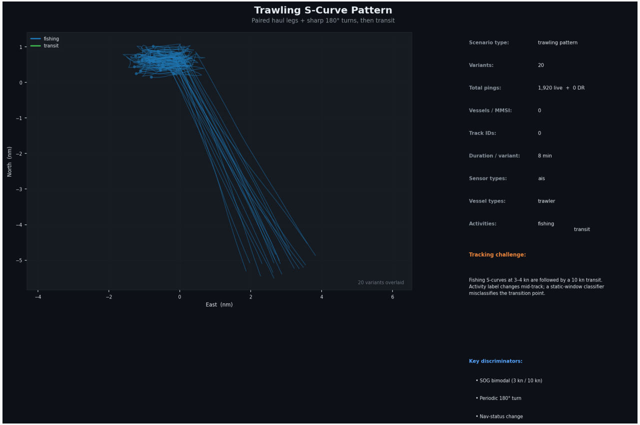
Why it's hard

The tracker must decide whether the reacquired contact is the same vessel (plausible at 4–8 kn over 4 h) or a new object. The uncertainty cone expands rapidly.

Ground truth

`is_dark=True` rows mark the

5 · Trawling Pattern



What happens

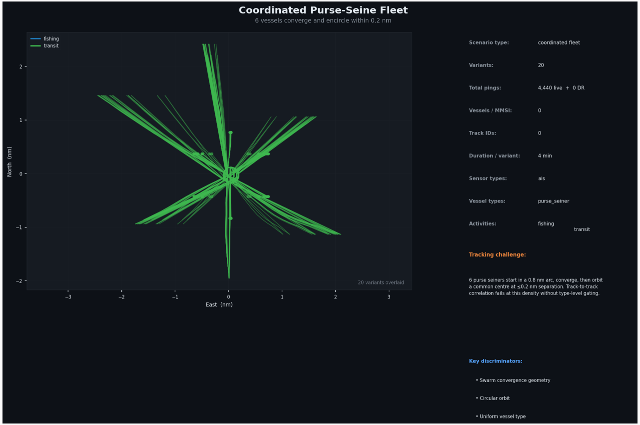
A trawler executes 5 paired haul-and-turn cycles (3–4 kn, 180° turns every ∼20 min) then transitions to a 10 kn transit to port.

Why it's hard

Activity changes mid-track. A static-window classifier labels the post-transition segment as fishing due to residual low-speed history.

Ground truth

6 · Coordinated Fishing Fleet



What happens

6 purse seiners start spread 0.8 nm apart, converge on a bait ball, then orbit a common centre at ≤ 0.2 nm separation.

Why it's hard

At orbit density, individual track gates overlap completely. Only vessel-type priors and staggered phase offsets allow disambiguation.

Ground truth

Each vessel has a distinct

7 · MMSI Cloning

What happens

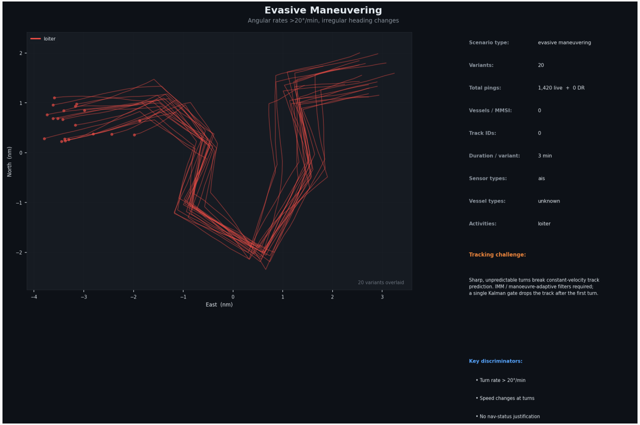
A fishing vessel clones a legitimate cargo vessel's MMSI. Both broadcast simultaneously from positions 15 nm apart.

Why it's hard

The data layer reports a single MMSI with two conflicting positions. Implied velocity between the two reports is physically impossible, which is the primary detection signal.

Ground truth

8 · Evasive Maneuvering



What happens

A vessel executes 6 sharp heading changes (>20°/min angular rate) in 70 minutes, with speed variations at each turn.

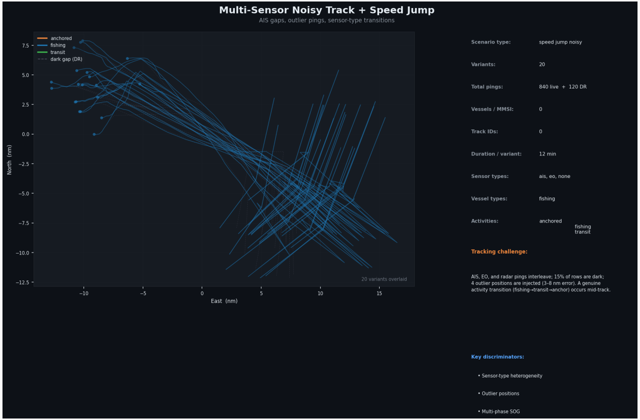
Why it's hard

A single-model Kalman filter loses the track after the first manoeuvre. IMM or manoeuvre-adaptive filters with wide process noise are required.

Ground truth

Activity is labelled `loiter`

9 · Multi-Sensor Noisy Track



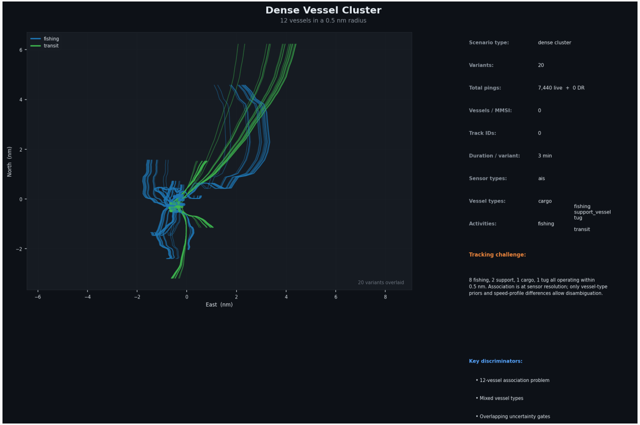
What happens

AIS, EO (satellite), and radar pings interleave. 15% of rows are dark. 4 outlier positions (3–8 nm error) are injected. The vessel transitions: `fishing` → `transit` → `anchored`.

Why it's hard

Sensor heterogeneity, position outliers, and missing data all coincide with genuine activity transitions, requiring robust outlier rejection and multi-modal fusion.

10 · Dense Vessel Cluster



What happens

12 vessels (8 fishing, 2 support, 1 cargo, 1 tug) operate within a 0.5 nm radius simultaneously.

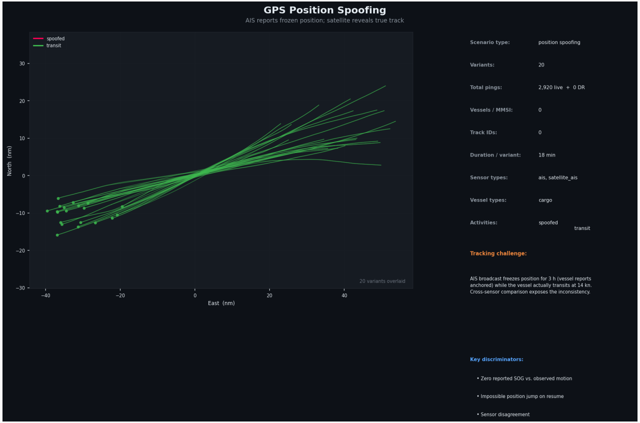
Why it's hard

Individual track gates overlap across the entire cluster for the full scenario duration. Only vessel-type priors and characteristic speed profiles allow any disambiguation.

Ground truth

Each vessel has a unique `mmsi`

11 · GPS Position Spoofing



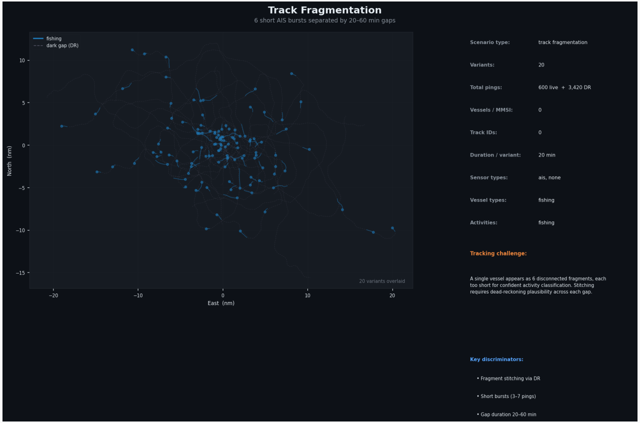
What happens

AIS broadcast freezes position for 3 h (reporting anchored, SOG≈0) while satellite-AIS reveals continuous transit at 14 kn.

Why it's hard

AIS-only systems classify the vessel as anchored for 3 h. The impossibly large position jump when honest AIS resumes is the retroactive detection signal. Cross-sensor fusion is required in

12 · Track Fragmentation



What happens

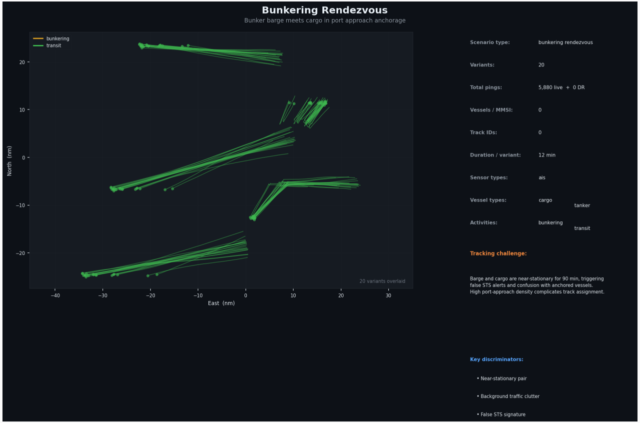
A single fishing vessel transmits in 6 short bursts (3–7 pings each) with 20–60 min dark gaps between them.

Why it's hard

Each fragment is individually too short for confident classification. Stitching the fragments into one track requires dead-reckoning plausibility checks across each gap.

Ground truth

13 · Bunkering Rendezvous



What happens

A bunker barge meets a Panamax cargo in a port-approach anchorage. The fuel transfer lasts 90 min. Three transiting vessels provide background clutter.

Why it's hard

The near-stationary pair triggers false STS alerts and is easily confused with anchored contacts. High traffic density in the anchorage complicates track assignment.

Geographic Variants — 20 Regions per Scenario

Each scenario type is instantiated in 20 maritime regions:

ID	Region	ID	Region
1	Singapore Strait (base)	11	North Sea
2	Malacca Strait (N)	12	Baltic Sea
3	South China Sea	13	Caribbean Sea
4	Gulf of Thailand	14	Gulf of Mexico
5	Bay of Bengal	15	US East Coast approaches
6	Arabian Sea	16	Pacific (Hawaii area)
7	Red Sea	17	Sea of Japan
8	Gulf of Aden	18	Korea Strait
9	West Africa / Gulf of Guinea	19	Taiwan Strait
10	Mediterranean (W)	20	Indian Ocean (central)

Each variant also applies an independent NumPy seed (1001–1020) and a speed scale factor (0.85–1.15) to vary traffic density and manoeuvre intensity.

How to Use This Dataset

Quick-start (Python) # Load one

```
scenario type
import pandas as pd
df = pd.read_csv(
  "crossing_tracks/variant_01.csv")

# Replay in time order
df = df.sort_values("timestamp")

# Split live vs dark rows
live = df[ df.is_dark]
dead_reck = df[df.is_dark]

# Load all 20 variants
df = pd.read_csv(
  "crossing_tracks_all_variants.csv")
```

Manifest

Evaluation notes

- `true_activity` is the ground truth — compare your classifier output against this column
- Filter `is_dark=True` rows when feeding a live classifier; they carry no real sensor observation
- `rng_seed` + `speed_scale` in the manifest allow exact reproduction of any variant
- The master file `all_scenarios.csv` contains all 39,340 rows with a `scenario` column for slicing

Technical Notes

Simulation model

Tracks are generated by dead-reckoning propagation with Gaussian course and speed noise. All positions are in WGS-84 decimal degrees. The spherical-Earth approximation is used (accuracy $<0.1\%$ at these scales).

Reproducibility

Every variant is seeded deterministically. Re-run `build_tracking_dataset.py` to regenerate identical files. The `rng_seed` column identifies the seed used.

Coordinate frame (figures)

Activity label set

- `fishing` — active gear deployment
- `transit` — underway at cruise speed
- `anchored` — stationary / moored
- `sts` — ship-to-ship transfer
- `bunkering` — fuel transfer
- `loiter` — slow / erratic, no defined activity
- `spoofed` — reported position known to be falsified

Sensor type codes